

Intro to Analysis

MATH 147

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§1 Course Information

§1.1 Logistics

§1.1.1 Contact

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Office hours: Monday, 3:30–4:30 p.m., and Friday, 10:00–11:00 a.m., in MC 4006.

§1.1.2 Course Structure

Lectures: Section 002 meets on Monday, Wednesday, and Friday from 12:30–1:20 p.m. in MC 4021.

§2 The Real Number System

Concept map. $\mathbb{R}$ is a complete ordered field, built in three layers: the field axioms (Theorem 2.3.1), the order axioms (Theorem 2.3.3), and completeness (Theorem 2.3.5, restated as the working axiom in §5). Completeness is the one axiom $\mathbb{Q}$ fails, and every later result traces back to it: the Archimedean property (Theorem 2.5.3) comes from completeness plus induction; it in turn drives the consequences in Theorem 2.5.4, the existence of $\sqrt{2}$ (Theorem 2.5.5), and the density of $\mathbb{Q}$ (Theorem 2.5.6). Suprema and infima (Theorems 2.4.2 and 2.4.3) with their ϵ -characterization (Theorem 2.4.4) are the tools that make these arguments run, and they return in §3 as the engine of the monotone convergence theorem.

§2.1 Numbers and Logic

§2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

§2.1.2 From $\mathbb{N}$ to $\mathbb{Q}$

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1.1 (Irrationality of $\sqrt{2}$)

The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.

In symbols

$$\neg(\exists m \in \mathbb{Z})(\exists n \in \mathbb{N}) \left[\sqrt{2} = \frac{m}{n} \right]$$

Idea

Write $\sqrt{2} = m/n$ in lowest terms. Then $m^2 = 2n^2$ forces m even, and feeding that back forces n even — so 2 divides both, contradicting lowest terms.

Proof. Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 2n^2.$$

Thus m^2 is even. An odd integer has an odd square, so m must be even. Write $m = 2\ell$

for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned} 2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2. \end{aligned}$$

Therefore n^2 is even, and the same parity argument shows that n is even.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

Assuming lowest terms turns the equation into a parity contradiction.

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.1.2 (Decimal representation)

Every nonnegative real number has a decimal representation of the form

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Negative real numbers are the negatives of such expansions. Some numbers have two decimal representations, for example $1.999\cdots = 2.000\cdots$.

Example 2.1.3 (Recurring decimal)

Is the statement $1.999\cdots = 2$ correct?

Yes. Let $x = 1.999\cdots$. Then

$$\begin{aligned} x &= 1.999\cdots, \\ 10x &= 19.999\cdots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

§2.1.3 Statements and Quantifiers

Definition 2.1.4 (Statement)

A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 2.1.5

The statement

$$(\forall x \in \mathbb{R}) [x^2 - x \geq 0]$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 2.1.6

The statement

$$(\exists x \in \mathbb{R}) [x^2 - x \geq 0]$$

is true; for instance, $x = 2$ satisfies the inequality.

Notation

Each quantifier is written in parentheses, and the statement it governs in square brackets, read left to right:

$$(\forall x \in \mathbb{R})(\exists y \in \mathbb{R}) [x^3 - y^3 = 1]$$

reads “for every $x \in \mathbb{R}$ there exists $y \in \mathbb{R}$ such that $x^3 - y^3 = 1$.” Negation moves inward and swaps each quantifier:

$$\neg(\forall x)(\exists y) [P(x, y)] \iff (\exists x)(\forall y) [\neg P(x, y)].$$

From here on, every theorem-like statement ends with its form *In symbols*: quantifiers and connectives only, using notation defined before that statement.

Proposition 2.1.7

For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that

$$x^3 - y^3 = 1.$$

In symbols

$$(\forall x \in \mathbb{R})(\exists y \in \mathbb{R}) [x^3 - y^3 = 1]$$

Proof. Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Here the choice of y is allowed to depend on x : for each x we produce a (possibly different) y that makes the equation hold. This is exactly what the quantifier order $(\forall x)(\exists y)$ permits.

Proposition 2.1.8

The statement

$$(\exists y \in \mathbb{R})(\forall x \in \mathbb{R}) [x^3 - y^3 = 1]$$

is false.

In symbols

$$\neg(\exists y \in \mathbb{R})(\forall x \in \mathbb{R}) [x^3 - y^3 = 1]$$

Its negation is

$$(\forall y \in \mathbb{R})(\exists x \in \mathbb{R}) [x^3 - y^3 \neq 1].$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark (Quantifier Order Matters). In this second statement, the quantifier order is reversed: $(\exists y)(\forall x)$ demands a *single* y that works simultaneously for every $x \in \mathbb{R}$. This is a much stronger requirement than $(\forall x)(\exists y)$, which only asks that a (possibly different) y exist for each individual x . Theorem 2.1.7 shows such y 's exist, but they genuinely depend on x (namely

$y = \sqrt[3]{x^3 - 1}$, so no single choice of y can work for all x at once. This is why swapping the order of $\forall$ and $\exists$ can change a true statement into a false one, and it is a caution to always read quantified statements in the order they are written.

§2.2 Induction and Well-Ordering

§2.2.1 Mathematical Induction

Definition 2.2.1 (Principle of Mathematical Induction)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true;

then

$$(\forall n \in \mathbb{N}) [P(n)]$$

is true.

In symbols

$$\left(P(1) \wedge (\forall k \in \mathbb{N}) [P(k) \Rightarrow P(k + 1)] \right) \Rightarrow (\forall n \in \mathbb{N}) [P(n)]$$

Here is an example:

Proposition 2.2.2 (Sum of the First n Natural Numbers)

For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}.$$

In symbols

$$(\forall n \in \mathbb{N}) \left[\sum_{k=1}^n k = \frac{n(n + 1)}{2} \right]$$

Proof. Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1 + 1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k + 1) &= \frac{k(k + 1)}{2} + (k + 1) \\ &= \frac{k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(k + 2)}{2}. \end{aligned}$$

Thus, $P(k + 1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 2.2.3

Let $S \subseteq \mathbb{N}$. If

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k + 1 \in S$;

then $S = \mathbb{N}$.

In symbols

$$(\forall S \subseteq \mathbb{N}) \left[(1 \in S \wedge (\forall k \in S) [k + 1 \in S]) \Rightarrow S = \mathbb{N} \right]$$

Proof. Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k + 1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

§2.2.2 The Well-Ordering Principle**Theorem 2.2.4 (Well-Ordering Principle)**

Every nonempty subset S of $\mathbb{N}$ has a smallest element.

In symbols

$$(\forall S \subseteq \mathbb{N}) \left[S \neq \emptyset \Rightarrow (\exists m \in S)(\forall s \in S) [m \leq s] \right]$$

Idea

If S had no smallest element, induction would show that S misses all of $1, 2, \dots, n$ for every n — so $S = \emptyset$.

Proof. Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

That is, T consists of exactly those $n \in \mathbb{N}$ for which none of $1, 2, \dots, n$ lies in S — informally, $n \in T$ means S has "not yet been seen" by the time we reach n . If we can show $T = \mathbb{N}$, then in particular no natural number lies in S (since every $n \in \mathbb{N}$ would satisfy $n \in T$, forcing $n \notin S$), giving $S = \emptyset$ and the desired contradiction. We show $T = \mathbb{N}$ by induction on n .

Base case. Since S has no smallest element and 1 is the smallest element of $\mathbb{N}$, we must have $1 \notin S$ (otherwise 1 would be the smallest element of S). Hence $\{1\} \cap S = \emptyset$, so $1 \in T$.

Inductive step. Suppose $k \in T$, i.e., $\{1, 2, \dots, k\} \cap S = \emptyset$. We show $k + 1 \in T$. Suppose, for contradiction, that $k + 1 \notin T$; then $\{1, 2, \dots, k + 1\} \cap S \neq \emptyset$. Since $\{1, \dots, k\} \cap S = \emptyset$ by the inductive hypothesis, the only way for $\{1, \dots, k + 1\} \cap S$ to be nonempty is if $k + 1 \in S$. But then $k + 1 \in S$ while none of $1, 2, \dots, k$ lie in S , so $k + 1$ would be the smallest element of S — contradicting our assumption that S has no smallest element. Hence $k + 1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so $S = \emptyset$, contradicting the assumption that S is nonempty. Therefore, S has a smallest element. ■

Proposition 2.2.5

Using the well-ordering principle, prove that for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

In symbols

$$(\forall n \in \mathbb{N}) \left[\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \right]$$

Idea

Minimal counterexample: let k be the first n where the formula fails. The formula still holds at $k-1$, and adding k^2 carries it to k .

Proof. Let $P(n)$ denote the statement

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Suppose, for contradiction, that $P(n)$ fails for some n ; that is,

$$\neg(\forall n \in \mathbb{N}) [P(n)] \quad \text{equivalently} \quad (\exists n \in \mathbb{N}) [\neg P(n)].$$

Then the set

$$S := \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By the well-ordering principle (Theorem 2.2.4), S has a smallest element k . Since $P(1)$ holds, $1 \notin S$, so $k > 1$ and hence $k-1 \in \mathbb{N}$.

By minimality of k , we have $k-1 \notin S$, so $P(k-1)$ is true:

$$1^2 + 2^2 + \cdots + (k-1)^2 = \frac{(k-1)k(2(k-1)+1)}{6} = \frac{(k-1)k(2k-1)}{6}.$$

Adding k^2 to both sides gives

$$\begin{aligned} 1^2 + \cdots + (k-1)^2 + k^2 &= \frac{(k-1)k(2k-1)}{6} + k^2 \\ &= \frac{(k-1)k(2k-1) + 6k^2}{6} \\ &= \frac{k((k-1)(2k-1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k+1)(2k+1)}{6}, \end{aligned}$$

so $P(k)$ is in fact true. This contradicts $k \in S$. Therefore $S = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$. ■

Remark (Why min, not sup). It is worth pausing on why we take the *minimum* of S here rather than some supremum-type argument. The well-ordering principle applies specifically to nonempty subsets of $\mathbb{N}$ (not to general subsets of $\mathbb{R}$), and it guarantees the existence of a smallest element that actually *belongs to* S — this is what allows us to write $k = \min S \in S$ and conclude $\neg P(k)$. This is a stronger statement than merely having an infimum: for a general subset of $\mathbb{R}$, an infimum need not belong to the set, so we could not conclude $\neg P(\inf S)$. Here,

because $S \subseteq \mathbb{N}$ and S is nonempty, well-ordering hands us an actual least element k with $\neg P(k)$ true, which is exactly what is needed to test P at k and at $k - 1$.

It is also worth noting that induction and the well-ordering principle are logically equivalent ways of expressing the same underlying fact about $\mathbb{N}$: the proof of the well-ordering principle above used induction, and here we use well-ordering to reprove a statement ($P(n)$ for all n) that could equally well have been proved by direct induction on n . The two techniques are simply different packagings of the same idea.

In practice

- **Induction.** To prove $P(n)$ for every $n \in \mathbb{N}$: verify the *base case* $P(1)$ directly; then fix an arbitrary $k \in \mathbb{N}$, assume $P(k)$ (the *inductive hypothesis*), and deduce $P(k + 1)$. No individual case needs to be checked beyond these two steps.
- **Minimal counterexample.** Suppose $P(n)$ fails somewhere. The counterexamples $S = \{n \in \mathbb{N} : \neg P(n)\}$ form a nonempty subset of $\mathbb{N}$, so they have a least element k . Because k is the *first* failure, $P(k - 1)$ still holds; show that $P(k - 1)$ forces $P(k)$:

$$S \neq \emptyset \Rightarrow k = \min S \Rightarrow P(k - 1) \Rightarrow P(k) \Rightarrow k \notin S, \text{ contradiction.}$$

§2.3 $\mathbb{R}$ as an Ordered Field

The real numbers $\mathbb{R}$ form a *complete ordered field*. We build up this description in three stages: first the field (algebraic) axioms, which govern $+$ and $\cdot$; then the order axioms, which let us compare elements with $<$; and finally the completeness axiom, which is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is the source of all limiting behaviour in analysis. Each stage adds structure on top of the previous one, and each is needed for a different purpose: the field axioms let us compute, the order axioms let us compare, and completeness lets us take limits.

§2.3.1 The Field Axioms

Definition 2.3.1 (Field)

A **field** is a set F equipped with two binary operations

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F,$$

satisfying the following axioms for all $a, b, c \in F$.

F1. Associativity of addition.

$$a + (b + c) = (a + b) + c.$$

F2. Commutativity of addition.

$$a + b = b + a.$$

F3. Additive identity.

There exists an element $0 \in F$ such that

$$a + 0 = a.$$

F4. Additive inverses.

For every $a \in F$, there exists an element $-a \in F$ such that

$$a + (-a) = 0.$$

F5. Associativity of multiplication.

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

F6. Commutativity of multiplication.

$$a \cdot b = b \cdot a.$$

F7. Multiplicative identity. There exists an element $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

F8. Multiplicative inverses. For every $a \in F$ with $a \neq 0$, there exists an element $a^{-1} \in F$ such that

$$a \cdot a^{-1} = 1.$$

F9. Distributive law.

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

The rational numbers $\mathbb{Q}$ also satisfy F1–F9, so being a field does not distinguish $\mathbb{R}$ from $\mathbb{Q}$: the field axioms capture the algebra of $+$ and $\cdot$, but nothing about size, order, or completeness.

Once the axioms are fixed, even the most familiar facts — uniqueness of the identities, $a \cdot 0 = 0$ — become theorems.

Theorem 2.3.2

- a) If $z, a \in \mathbb{R}$ with $z + a = a$, then $z = 0$.
- b) If $u \in \mathbb{R}$ and $b \neq 0$ with $u \cdot b = b$, then $u = 1$.
- c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

In symbols

- a) $(\forall z, a \in \mathbb{R}) [z + a = a \Rightarrow z = 0]$
- b) $(\forall u, b \in \mathbb{R}) [(b \neq 0 \wedge u \cdot b = b) \Rightarrow u = 1]$
- c) $(\forall a \in \mathbb{R}) [a \cdot 0 = 0]$

Proof. a)

$$z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0.$$

b) Multiplying $u \cdot b = b$ by b^{-1} ,

$$u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1.$$

c)

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides, and using associativity, gives $0 = a \cdot 0$. ■

§2.3.2 The Order Axioms**Definition 2.3.3 (The Order Properties of $\mathbb{R}$)**

There is a nonempty subset P of $\mathbb{R}$, called the set of **positive real numbers**, that satisfies the following properties.

O1. Trichotomy law. For every $a \in \mathbb{R}$, exactly one of the following holds:

$$a = 0, \quad a \in P, \quad -a \in P.$$

O2. Closure under addition. If $a, b \in P$, then $a + b \in P$.

O3. Closure under multiplication. If $a, b \in P$, then $a \cdot b \in P$.

Definition 2.3.4 (Order relation)

Let $a, b \in \mathbb{R}$.

- (i) If $a - b \in P$, then we write $a > b$ or $b < a$.
- (ii) If $a - b \in P \cup \{0\}$, then we write $a \geq b$ or $b \leq a$.

The positive rationals $P \cap \mathbb{Q}$ also satisfy O1–O3, so an ordered field still need not be $\mathbb{R}$. The missing ingredient is completeness, which $\mathbb{Q}$ lacks: the set

$$\{q \in \mathbb{Q} : q^2 < 2\}$$

is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$. That is why completeness is a separate axiom.

Definition 2.3.5 (Completeness Axiom)

If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

In symbols

$$\begin{aligned} (\forall A \subseteq \mathbb{R}) \Big[& (A \neq \emptyset \wedge (\exists u \in \mathbb{R})(\forall a \in A) [a \leq u]) \\ & \Rightarrow (\exists u_0 \in \mathbb{R}) [(\forall a \in A) [a \leq u_0] \\ & \wedge (\forall u \in \mathbb{R}) ((\forall a \in A) [a \leq u] \Rightarrow u_0 \leq u)] \Big] \end{aligned}$$

This is only a preview: upper bounds, suprema and the consequences of this axiom are developed in §5.

Example 2.3.6

Prove that if $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof. Since $a \neq 0$, the trichotomy law gives two cases.

Case 1: $a \in P$. By closure under multiplication,

$$a^2 = a \cdot a \in P, \quad \text{so } a^2 > 0.$$

Case 2: $a \notin P$. Since $a \neq 0$, trichotomy rules out $a = 0$, so it must be that $-a \in P$. By closure under multiplication, $(-a)(-a) \in P$. It remains to show that $a^2 = (-a)(-a)$, which we do by starting from $a + (-a) = 0$ and multiplying both sides by $(-a)$:

$$\begin{aligned} a + (-a) = 0 & \implies (-a)(a + (-a)) = (-a) \cdot 0 = 0 \\ & \implies (-a)(a) + (-a)(-a) = 0 \\ & \implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

Hence $a^2 = (-a)(-a) \in P$, so $a^2 > 0$. Since the two cases $a \in P$ and $-a \in P$ are exhaustive (trichotomy, together with $a \neq 0$), we conclude $a^2 > 0$ in either case. $\blacksquare$

§2.3.3 Absolute Value and the Triangle Inequality

Definition 2.3.7

Given $a \in \mathbb{R}$, we define the **absolute value** $|a|$ by

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Geometrically, $|a|$ is the distance from a to 0, and $|a - b|$ is the distance between a and b ; the results below are statements about distances.

Theorem 2.3.8

(a) $|ab| = |a| |b|$ for all $a, b \in \mathbb{R}$.

(b) $|a|^2 = a^2$ for all $a \in \mathbb{R}$.

(c) For $a \in \mathbb{R}$ and $c > 0$,

$$|a| \leq c \iff -c \leq a \leq c.$$

(d) For every $a \in \mathbb{R}$,

$$-|a| \leq a \leq |a|.$$

In symbols

$$(a) \quad (\forall a, b \in \mathbb{R}) [|ab| = |a| |b|]$$

$$(b) \quad (\forall a \in \mathbb{R}) [|a|^2 = a^2]$$

$$(c) \quad (\forall a \in \mathbb{R})(\forall c > 0) [|a| \leq c \iff -c \leq a \leq c]$$

$$(d) \quad (\forall a \in \mathbb{R}) [-|a| \leq a \leq |a|]$$

Idea

Every part is the same move: split on the sign of a (and of b), and unwind the definition of $|\cdot|$ in each case.

Proof. The definition immediately gives $|-a| = |a|$: if $a \geq 0$, then $-a \leq 0$, and if $a < 0$, then $-a > 0$. It also gives

$$-|a| \leq a \leq |a|,$$

proving (d). In the two cases $a \geq 0$ and $a < 0$, respectively, $|a|^2 = a^2$ is immediate; this proves (b).

For (a), there are four sign combinations. If $a, b \geq 0$, then $ab \geq 0$, so $|ab| = ab = |a| |b|$. If $a \geq 0 > b$, then $ab \leq 0$, so $|ab| = -ab = a(-b) = |a| |b|$. The case $b \geq 0 > a$ is the same with the roles reversed. Finally, if $a, b < 0$, then $ab > 0$, and

$$|ab| = ab = (-a)(-b) = |a| |b|.$$

Finally let $c > 0$. Suppose first that $|a| \leq c$. If $a \geq 0$, then $|a| = a$, so $a \leq c$; also $-c < 0 \leq a$. If $a < 0$, then $|a| = -a$, so $-a \leq c$, equivalently $-c \leq a$; also $a < 0 < c$. Hence $-c \leq a \leq c$ in either case.

Conversely, suppose $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$. If $a < 0$, multiplying $-c \leq a$ by -1 reverses the inequality and gives $-a \leq c$, hence $|a| \leq c$. This proves (c). ■

Proposition 2.3.9 (Triangle Inequality)

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

In symbols

$$(\forall a, b \in \mathbb{R}) [|a + b| \leq |a| + |b|]$$

Idea

Add the two-sided bounds from part (d) for a and for b ; this gives $-c \leq a + b \leq c$ with $c = |a| + |b|$, and part (c) turns that back into $|a + b| \leq c$.

Proof. Let $a, b \in \mathbb{R}$. By part (d) of Theorem 2.3.8,

$$\begin{aligned} -|a| &\leq a \leq |a|, \\ -|b| &\leq b \leq |b|. \end{aligned}$$

Adding these two chains of inequalities termwise gives

$$\begin{aligned} -|a| - |b| &\leq a + b \leq |a| + |b| \\ \iff -(|a| + |b|) &\leq a + b \leq |a| + |b|. \end{aligned}$$

If $|a| + |b| = 0$, then the displayed bounds force $a + b = 0$, so the claim is immediate. Otherwise $|a| + |b| > 0$, and applying part (c) of Theorem 2.3.8 with $x = a + b$ and $c = |a| + |b|$ converts this two-sided bound back into the single inequality

$$|a + b| \leq |a| + |b|.$$

■

In terms of distance: going from 0 to $a + b$ directly is never longer than going via a .

Corollary 2.3.10

- (i) $||a| - |b|| \leq |a - b|$.
- (ii) $|a - b| \leq |a| + |b|$

In symbols

- (i) $(\forall a, b \in \mathbb{R}) [|a| - |b| \leq |a - b|]$
- (ii) $(\forall a, b \in \mathbb{R}) [|a - b| \leq |a| + |b|]$

Proof of (i). Let $a, b \in \mathbb{R}$. The idea is to write a as $(a - b) + b$, so that the triangle inequality just proved applies directly to this sum and produces a bound involving $|a - b|$:

$$\begin{aligned} a &= (a - b) + b \implies |a| = |(a - b) + b| \leq |a - b| + |b| \\ &\implies |a| - |b| \leq |a - b|. \end{aligned}$$

Symmetrically, writing b as $(b - a) + a$ and applying the triangle inequality the same way

gives

$$\begin{aligned} b &= (b - a) + a \implies |b| = |(b - a) + a| \leq |b - a| + |a| \\ &\implies |b| - |a| \leq |b - a| = |-(a - b)| = |a - b| \\ &\implies |a| - |b| \geq -|a - b|. \end{aligned}$$

Combining the two bounds obtained from these two decompositions,

$$-|a - b| \leq |a| - |b| \leq |a - b|.$$

By part (c) of Theorem 2.3.8, applied with $x = |a| - |b|$ and $c = |a - b|$, this two-sided bound is equivalent to

$$||a| - |b|| \leq |a - b|.$$

■

Proof of (ii). Let $a, b \in \mathbb{R}$. Writing $a - b$ as $a + (-b)$ reduces this to a direct application of the triangle inequality, using $|-b| = |b|$:

$$|a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

■

In practice

- **Algebra is F1–F9.** Every “obvious” manipulation is an axiom in disguise: moving a term across “=” uses additive inverses and associativity, cancelling a nonzero factor uses multiplicative inverses, expanding a product uses the distributive law.
- **Case-split with trichotomy.** Any $a \in \mathbb{R}$ falls in exactly one of $a = 0$, $a \in P$, $-a \in P$; handle each case, and use O2, O3 to keep positivity through the algebra (as in the proof that $a \neq 0 \Rightarrow a^2 > 0$).
- **To show $a > b$, show $a - b \in P$.** Every order manipulation (adding to both sides, multiplying by a positive number) traces back to this definition and O1–O3.
- **Unpack absolute values.** $|x| \leq c \iff -c \leq x \leq c$ turns one absolute-value inequality into two ordinary ones, and back. The triangle inequality is the first example; it will be used constantly.
- **Add and subtract.** To bound $|a|$ in terms of $|a - b|$, write $a = (a - b) + b$ and apply the triangle inequality.

§2.4 Suprema and Infima

Bounds make sense in any ordered set. The special feature of $\mathbb{R}$ is that every nonempty set bounded above has a supremum in $\mathbb{R}$; this need not hold in $\mathbb{Q}$.

Definition 2.4.1 (Bounded Sets)

Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be

- (i) bounded above if there exists $u \in \mathbb{R}$ such that $x \leq u$ for all $x \in S$.
- (ii) bounded below if there exists $w \in \mathbb{R}$ such that $w \leq x$ for all $x \in S$.
- (iii) bounded if it is both bounded above and below.
- (iv) unbounded if it is not bounded.

Definition 2.4.2 (Supremum)

If there exists an upper bound u_0 of S such that whenever u is an upper bound for S we have $u_0 \leq u$, then u_0 is called the least upper bound, or the supremum, of S .

$$u_0 := \sup S.$$

Definition 2.4.3 (Infimum)

If there exists a lower bound w_0 of S such that whenever w is a lower bound for S we have $w_0 \geq w$, then w_0 is called the greatest lower bound, or the infimum, of S .

$$w_0 := \inf S.$$

A set has at most one supremum. If u_1 and u_2 are both suprema of S , then u_1 is least and u_2 is an upper bound, so $u_1 \leq u_2$; swapping roles gives $u_2 \leq u_1$. Hence $u_1 = u_2$, and the same argument with inequalities reversed shows the infimum is unique.

§2.4.1 The ϵ -Characterization of the Supremum

When does an upper bound u fail to be the *least* one? Exactly when some smaller number is still an upper bound — that is, when there is a gap between S and u . So u is the supremum precisely when there is no gap: however small a margin $\epsilon > 0$ we allow, S still has a point within ϵ of u .

Lemma 2.4.4

An upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S) [u - \epsilon < x_\epsilon].$$

In symbols

$$\begin{aligned} & (\forall S \subseteq \mathbb{R})(\forall u \in \mathbb{R}) \left[(S \neq \emptyset \wedge (\forall x \in S) [x \leq u]) \right. \\ & \quad \left. \Rightarrow (u = \sup S \iff (\forall \epsilon > 0)(\exists x_\epsilon \in S) [u - \epsilon < x_\epsilon]) \right] \end{aligned}$$

Proof. ($\Rightarrow$) Suppose $u = \sup S$. Let $\epsilon > 0$. Since

$$u - \epsilon < u,$$

the number $u - \epsilon$ cannot be an upper bound: otherwise it would be an upper bound smaller than u , contradicting that u is the *least* upper bound. Since $u - \epsilon$ is not an upper bound, by the definition of upper bound there is some $x_\epsilon \in S$ such that

$$u - \epsilon < x_\epsilon \leq u.$$

($\Leftarrow$) Conversely, suppose u is an upper bound and that for every $\epsilon > 0$ there exists $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$. If u were not the least upper bound, there would be another upper bound $w < u$. Choose

$$\epsilon = u - w > 0.$$

The assumed condition then yields some $x_\epsilon \in S$ with

$$w = u - \epsilon < x_\epsilon,$$

contradicting that w is an upper bound (since $x_\epsilon \in S$ exceeds w). Hence no such w exists, and $u = \sup S$. ■

Example 2.4.5

Consider the set $S = \{x \in \mathbb{R} : 0 \leq x < 1\}$. Then $\sup S = 1$ and $\inf S = 0$.

Proof that $\sup S = 1$. Every $x \in S$ satisfies $x < 1$, so 1 is an upper bound. Let $u = \sup S$, which exists by completeness since S is nonempty and bounded above. Because 1 is an upper bound, $u \leq 1$. Suppose $u \neq 1$, so $u < 1$. Choose

$$\epsilon = \frac{1 - u}{2} > 0.$$

Then

$$u < u + \epsilon = \frac{u + 1}{2} < 1.$$

Also $u \geq 0$ since $0 \in S$, so $u + \epsilon \geq 0$. Thus $u + \epsilon \in S$, contradicting that u is an upper bound for S . Therefore $u = 1$, i.e. $\sup S = 1$. ■

Proof that $\inf S = 0$. Since $x \geq 0$ for all $x \in S$, 0 is a lower bound. Let $w = \inf S$. Because 0 is a lower bound, $w \geq 0$. Suppose $w \neq 0$, so $w > 0$. But $0 \in S$, and a lower bound w must satisfy $w \leq 0$, a contradiction. Therefore $w = 0$, i.e. $\inf S = 0$. ■

Remark (Supremum versus Maximum). In this example, $S = [0, 1)$ has $\sup S = 1$ but no maximum, because $1 \notin S$. Its infimum 0 does belong to S , so 0 is also its minimum. A supremum is the *least upper bound*, whether or not it belongs to the set; it is a maximum only when it does belong to the set. Similarly, “bounded above,” “bounded below,” “bounded,” and “unbounded” are properties of the *set* S , whereas $\sup S$ and $\inf S$ (when they exist) are specific *real numbers* associated to S .

In practice

- **Verifying $u = \sup S$.** Check two things: (i) u is an upper bound for S ; (ii) no smaller number is. Condition (ii) quantifies over all numbers below u and is awkward directly; use the ϵ -form instead: for each $\epsilon > 0$, exhibit $x \in S$ with $u - \epsilon < x \leq u$.
- **Infimum.** The same recipe with inequalities reversed: a lower bound w , and for each $\epsilon > 0$ some $x \in S$ with $w \leq x < w + \epsilon$.

§2.5 Completeness and Its Consequences

Axiom (Completeness Property)

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

In symbols

$$\begin{aligned} (\forall S \subseteq \mathbb{R}) \left[(S \neq \emptyset \wedge (\exists u \in \mathbb{R})(\forall x \in S) [x \leq u]) \right. \\ \left. \Rightarrow (\exists u_0 \in \mathbb{R}) [u_0 = \sup S] \right] \end{aligned}$$

Corollary 2.5.1

Every nonempty set of real numbers that has a lower bound also has an infimum in $\mathbb{R}$.

In symbols

$$\begin{aligned} & (\forall S \subseteq \mathbb{R}) \left[(S \neq \emptyset \wedge (\exists w \in \mathbb{R})(\forall x \in S) [w \leq x]) \right. \\ & \quad \left. \Rightarrow (\exists w_0 \in \mathbb{R}) [w_0 = \inf S] \right] \end{aligned}$$

Proof. Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , that is, $w \leq s$ for all $s \in S$. As a consequence, $-w \geq -s$ for all $s \in S$.

Define $S' = \{-s : s \in S\}$. Then S' is nonempty, and

$$(\forall s \in S) [-w \geq -s] \implies -w \text{ is an upper bound for } S'.$$

By the Completeness Axiom, S' has a supremum. Say

$$u := \sup S' \in \mathbb{R}.$$

We claim that $-u = \inf S$.

- **Lower bound:** Since $u = \sup S' \geq -s$ for all $s \in S$, we get $-u \leq s$ for all $s \in S$. So $-u$ is a lower bound for S .
- **Greatest lower bound:** Let w be any lower bound for S . As shown above, $-w$ is then an upper bound for S' , so $u = \sup S' \leq -w$, which gives $w \leq -u$.

Thus $-u$ is a lower bound for S that is at least as large as every other lower bound, so $-u = \inf S$. ■

Example 2.5.2

$$S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$$

For every $n \in \mathbb{N}$, the binomial theorem gives

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \sum_{k=2}^n \binom{n}{k} \frac{1}{n^k} > 1.$$

Indeed, for $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}.$$

For $k \geq 2$, we have $k! \geq 2^{k-1}$, because each factor from 2 through k is at least 2. Therefore

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

Thus $1 < (1 + 1/n)^n < 3$ for all $n \in \mathbb{N}$.

§2.5.1 The Archimedean Property

We now prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. The proof combines two ingredients:

- Completeness of $\mathbb{R}$: this is what lets us form $u = \sup \mathbb{N}$ in the first place. Note carefully that u here is the supremum of $\mathbb{N}$ *regarded as a subset of* $\mathbb{R}$ — completeness says nothing about whether u itself is a natural number, and indeed the entire argument below is

designed to show that assuming $\mathbb{N}$ is bounded above (so that such a u exists) leads to a contradiction.

- The inductive property of $\mathbb{N}$ (that $\mathbb{N}$ is closed under $+1$): this is the purely arithmetic fact that lets us manufacture a natural number strictly larger than u , which is what breaks the supremum.

Definition 2.5.3 (Archimedean Property)

If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.

In symbols

$$(\forall x \in \mathbb{R})(\exists n_x \in \mathbb{N}) [x \leq n_x]$$

Idea

If $\mathbb{N}$ were bounded above, let $u = \sup \mathbb{N}$. Then $u - 1$ is not an upper bound (the ϵ -characterization with $\epsilon = 1$), so some $m \in \mathbb{N}$ has $m > u - 1$ — and then $m + 1 \in \mathbb{N}$ lies beyond u .

Proof. Suppose, for contradiction, that $(\exists x \in \mathbb{R})(\forall n \in \mathbb{N}) [x > n]$.

Then x is an upper bound for $\mathbb{N}$. Since $1 \in \mathbb{N}$, $\mathbb{N} \neq \emptyset$, so by the Completeness Axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$.

Say $u = \sup \mathbb{N}$. Since

$$u - 1 < u,$$

and u is the *least* upper bound of $\mathbb{N}$, no number strictly smaller than u can be an upper bound for $\mathbb{N}$ (an upper bound smaller than the least upper bound would contradict its minimality). Applying this general principle to $u - 1$, we conclude that $u - 1$ is not an upper bound for $\mathbb{N}$. Hence

$$(\exists m \in \mathbb{N}) [u - 1 < m],$$

which gives

$$u < m + 1.$$

But $m + 1 \in \mathbb{N}$ (since $\mathbb{N}$ is closed under $+1$), and u is an upper bound for $\mathbb{N}$, so $m + 1 \leq u$. This contradicts $u < m + 1$: we have simultaneously $u < m + 1$ and $m + 1 \leq u$.

Therefore no such x exists; that is, for every $x \in \mathbb{R}$ there exists $n_x \in \mathbb{N}$ with $x \leq n_x$. ■

Completeness + closure of $\mathbb{N}$ under $+1 \implies$ Archimedean Property

Corollary 2.5.4

The following are consequences of the Archimedean Property:

- (a) If

$$S = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\},$$

then

$$\inf S = 0.$$

- (b) If $t > 0$, there exists $n_t \in \mathbb{N}$ such that

$$0 < \frac{1}{n_t} < t.$$

(c) If $y > 0$, there exists $n_y \in \mathbb{N}$ such that

$$n_y - 1 \leq y < n_y.$$

In symbols

- (a) $\inf \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} = 0$
- (b) $(\forall t > 0)(\exists n_t \in \mathbb{N}) \left[0 < \frac{1}{n_t} < t \right]$
- (c) $(\forall y > 0)(\exists n_y \in \mathbb{N}) [n_y - 1 \leq y < n_y]$

Idea

Each part is one move.

- (a) 0 is a lower bound, and (b) says nothing larger is.
- (b) Archimedes applied to $1/t$.
- (c) Take the *least* natural number to the right of y : $n_y = \min \{n \in \mathbb{N} : y < n\}$ (well-ordering, not a supremum).

Proof. (a) Clearly 0 is a lower bound for S . Let $b > 0$. By part (b), there exists $n \in \mathbb{N}$ such that $1/n < b$. Since $1/n \in S$, this means b is not a lower bound for S . As $b > 0$ was arbitrary, no number greater than 0 is a lower bound for S . Combined with 0 being a lower bound, $\inf S = 0$.

(b) By the Archimedean Property, choose $m \in \mathbb{N}$ with $1/t \leq m$. Then $n_t = m + 1 \in \mathbb{N}$ satisfies $n_t > m \geq 1/t$, so $n_t > 1/t$, and dividing by $n_t > 0$ gives $0 < 1/n_t < t$.

(c) Let

$$A = \{n \in \mathbb{N} : y < n\}.$$

By the Archimedean Property, A is nonempty (some natural number exceeds y). Since $A \subseteq \mathbb{N}$ is nonempty, the Well-Ordering Principle guarantees that A has a least element; let $n_y = \min A$. Then $n_y \in A$, so $y < n_y$.

It remains to show $n_y - 1 \leq y$. Suppose instead that $n_y - 1 > y$. Since $y > 0$ and $n_y \geq 1$ (as $n_y \in \mathbb{N}$), if $n_y = 1$ then $n_y - 1 = 0 > y$ is impossible because $y > 0$; so $n_y \geq 2$ and $n_y - 1 \in \mathbb{N}$. Then $n_y - 1 \in \mathbb{N}$ together with $n_y - 1 > y$ shows $n_y - 1 \in A$. But $n_y - 1 < n_y$, contradicting that n_y is the *least* element of A . Hence $n_y - 1 \leq y$, and combined with $y < n_y$ we obtain

$$n_y - 1 \leq y < n_y.$$

■

Remark (Minimum versus Supremum). Part (c) is a *minimum* argument, not a supremum argument, and this distinction matters. The set $A = \{n \in \mathbb{N} : y < n\}$ is a subset of $\mathbb{N}$, and in general A is *unbounded above* in $\mathbb{R}$: if y is not itself a boundary case, every sufficiently large natural number also lies in A . So $\sup A$ would not even be a finite real number, and asking for $\sup A$ would be the wrong tool here. What we actually need is the *least* element of A , and the Well-Ordering Principle supplies this regardless of whether A is bounded above — it only requires A to be a nonempty subset of $\mathbb{N}$. This is in direct contrast to the supremum arguments used elsewhere in this section (for $\sup \mathbb{N}$ in the Archimedean Property, and for $\sup S$ in the existence of $\sqrt{2}$ below), where the relevant sets are subsets of $\mathbb{R}$ and boundedness above is essential.

§2.5.2 The Existence of $\sqrt{2}$

We know $\sqrt{2} \notin \mathbb{Q}$. Does $\mathbb{R}$ contain a square root of 2? Completeness should produce one. Take all the candidates from below,

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

Its supremum $x = \sup S$ ought to sit exactly on the boundary between squares below 2 and squares above 2. There are only two ways x could miss:

- $x^2 < 2$: then x has not reached the boundary, so it can be nudged slightly *right* and still lie in S — then x is not an upper bound;
- $x^2 > 2$: then x has overshoot, so it can be nudged slightly *left* and still be an upper bound — then x is not the *least* one.

Unlike part (c) above, this is a genuine completeness argument: S is a bounded set of reals and we need its least upper bound, not a least element.

Proposition 2.5.5

There exists a real number x such that $x^2 = 2$.

In symbols

$$(\exists x \in \mathbb{R}) [x^2 = 2]$$

Scratch work

How small must the nudge $1/n$ be? Work backwards from what we want.

- *Right nudge.* Want $(x + \frac{1}{n})^2 < 2$. Expanding gives $x^2 + \frac{2x}{n} + \frac{1}{n^2}$; the $\frac{1}{n^2}$ is awkward, but $\frac{1}{n^2} \leq \frac{1}{n}$, so it is enough that

$$x^2 + \frac{2x+1}{n} < 2 \iff n > \frac{2x+1}{2-x^2}.$$

- *Left nudge.* Want $(x - \frac{1}{n})^2 > 2$. Dropping the positive $\frac{1}{n^2}$, it is enough that

$$x^2 - \frac{2x}{n} > 2 \iff n > \frac{2x}{x^2-2}.$$

Both thresholds are fixed positive numbers, so the Archimedean Property supplies such an n .

Proof. Define

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

We have $1 \in S$, so $S \neq \emptyset$, and 2 is an upper bound for S : if $t \in S$ and $t > 2$, then $t^2 > 4 > 2$, a contradiction. By the Completeness Axiom, S has a supremum in $\mathbb{R}$.

Let $x = \sup S$. Note $x \geq 1 > 0$. We show $x^2 = 2$ by ruling out $x^2 < 2$ and $x^2 > 2$.

Case 1: $x^2 < 2$. We find $n \in \mathbb{N}$ large enough that $x + \frac{1}{n} \in S$, contradicting that x is an upper bound for S .

Step 1 (bound the perturbed square). Since $x \geq 1$ and $n \geq 1$,

$$\begin{aligned} \left(x + \frac{1}{n}\right)^2 &= x^2 + \frac{2x}{n} + \frac{1}{n^2} \\ &\leq x^2 + \frac{2x}{n} + \frac{1}{n} \\ &= x^2 + \frac{2x+1}{n}. \end{aligned}$$

Step 2 (choose n via the Archimedean Property). Since $x^2 < 2$, the quantity $2 - x^2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x+1}{n} < 2 - x^2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x + \frac{1}{n}\right)^2 < x^2 + (2 - x^2) = 2,$$

so $x + \frac{1}{n} \in S$. But $x + \frac{1}{n} > x = \sup S$, contradicting that x is an upper bound for S . Hence $x^2 \geq 2$.

$x^2 < 2 \implies x$ can be moved slightly to the right
and remain in $S \implies x \neq \sup S$.

Case 2: $x^2 > 2$. We find $n \in \mathbb{N}$ large enough that $x - \frac{1}{n}$ is an upper bound for S smaller than x , contradicting that $x = \sup S$.

Step 1 (bound the perturbed square). Since $1/n^2 > 0$,

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n}.$$

Step 2 (choose n via the Archimedean Property). Since $x^2 > 2$, the quantity $x^2 - 2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x}{n} < x^2 - 2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x - \frac{1}{n}\right)^2 > x^2 - (x^2 - 2) = 2.$$

So for any $t \in S$, we have $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$. Also $x - 1/n \geq 0$ because $x \geq 1$ and $n \geq 1$. Since $u \mapsto u^2$ is strictly increasing on $[0, \infty)$, this implies $t < x - \frac{1}{n}$. Thus $x - \frac{1}{n}$ is an upper bound for S smaller than $x = \sup S$, contradicting that x is the *least* upper bound. Hence $x^2 \leq 2$.

$x^2 > 2 \implies x$ can be moved slightly to the left
and remain an upper bound for $S \implies x \neq \sup S$.

Combining both cases, $x^2 = 2$. ■

§2.5.3 Density of $\mathbb{Q}$ and of the Irrationals

Completeness is exactly what $\mathbb{Q}$ is missing. Take

$$T = \{r \in \mathbb{Q} : r^2 < 2\},$$

the rational version of the set S above. It is nonempty ($0 \in T$) and bounded above in $\mathbb{Q}$ (every $r \in T$ has $r < 2$). Viewed inside $\mathbb{R}$, it has a supremum $y = \sup_{\mathbb{R}} T$, and the same two-case argument as above gives $y^2 = 2$, so $y = \sqrt{2}$. But $\sqrt{2} \notin \mathbb{Q}$, and y was the only possible least upper bound, so T has no supremum in $\mathbb{Q}$. Thus $\mathbb{Q}$ is *not* complete: $\mathbb{R}$ needs irrationals like $\sqrt{2}$ to serve as suprema of sets like T .

Still, $\mathbb{Q}$ is spread densely through $\mathbb{R}$. The picture: lay down the multiples $\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots$ on the number line. If the step $\frac{1}{n}$ is shorter than the interval (x, y) , the multiples cannot jump over it, so one of them lands inside.

Theorem 2.5.6

If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

In symbols

$$(\forall x, y \in \mathbb{R}) [x < y \Rightarrow (\exists r \in \mathbb{Q}) [x < r < y]]$$

Scratch work

Take $0 < x < y$ first. We want m, n with

$$x < \frac{m}{n} < y \iff xn < m < yn.$$

- Choose m to be the *first* natural number past xn (part (c)): then $xn < m \leq xn + 1$.
- This m stays below yn provided $xn + 1 < yn$, i.e. $\frac{1}{n} < y - x$ — which Archimedes provides.

For $x \leq 0$, shift both endpoints right by an integer $k > -x$, apply the positive case, and shift back.

Proof. First suppose that $0 < x < y$. Then $y - x > 0$. By the Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < y - x.$$

Multiplying $1/n < y - x$ by $n > 0$ shows that

$$xn + 1 < yn.$$

We now find a natural number between xn and $xn + 1$. Since $xn > 0$, part (c) of Theorem 2.5.4 (with y there replaced by xn) gives some $m \in \mathbb{N}$ with

$$m - 1 \leq xn < m.$$

Equivalently,

$$xn < m \leq xn + 1.$$

Combining this with $xn + 1 < yn$ obtained above, we have

$$xn < m < yn,$$

where the left inequality is immediate from $m > xn$, and the right inequality follows since $m \leq xn + 1 < yn$. Dividing throughout by $n > 0$, we get

$$x < \frac{m}{n} < y.$$

Since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, $m/n \in \mathbb{Q}$. This proves the theorem when $0 < x < y$.

Now suppose that $x \leq 0$. Choose $k \in \mathbb{N}$ such that $-x < k$ (possible by the Archimedean Property). Then adding k to $x < y$ and to $-x < k$ gives

$$0 < x + k < y + k.$$

By the positive case just proved, there exists $q \in \mathbb{Q}$ such that $x + k < q < y + k$. Let $r = q - k$. Since $q \in \mathbb{Q}$ and $k \in \mathbb{N} \subseteq \mathbb{Q}$, we have $r \in \mathbb{Q}$, and subtracting k from each part of $x + k < q < y + k$ gives

$$x < r < y.$$

■

Corollary 2.5.7

If x and y are any real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.

In symbols

$$(\forall x, y \in \mathbb{R}) [x < y \Rightarrow (\exists z \in \mathbb{R} \setminus \mathbb{Q}) [x < z < y]]$$

Idea

Scale by $\sqrt{2}$: a nonzero rational r between $x/\sqrt{2}$ and $y/\sqrt{2}$ gives $\sqrt{2}r$ between x and y , and $\sqrt{2}r$ is irrational.

Proof. Set $a = x/\sqrt{2}$ and $b = y/\sqrt{2}$. Since $\sqrt{2} > 0$, we have $a < b$. Choose a nonzero rational number r with $a < r < b$: if $b \leq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; if $a \geq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; and if $a < 0 < b$, choose $r \in \mathbb{Q}$ with $0 < r < b$. In each case, Theorem 2.5.6 provides such an r , and $r \neq 0$.

Let $z = \sqrt{2}r$. Multiplying $a < r < b$ by $\sqrt{2}$ gives $x < z < y$. If z were rational, then $\sqrt{2} = z/r$ would be rational, a contradiction. Thus z is irrational. ■

Corollary 2.5.8

Let $x, y \in \mathbb{R}$, with $x < y$. Then there exists infinitely many rationals between x and y .

In symbols

$$\begin{aligned} &(\forall x, y \in \mathbb{R}) [x < y \\ &\Rightarrow (\forall n \in \mathbb{N})(\exists r_1, \dots, r_n \in \mathbb{Q}) [x < r_1 < r_2 < \dots < r_n < y]] \end{aligned}$$

Proof. Choose $r_1 \in \mathbb{Q}$ with $x < r_1 < y$. Having chosen $r_k \in \mathbb{Q}$ with $x < r_k < y$, apply Theorem 2.5.6 to $r_k < y$ to choose $r_{k+1} \in \mathbb{Q}$ such that

$$r_k < r_{k+1} < y.$$

This produces infinitely many distinct rational numbers $r_1, r_2, \dots$ in (x, y) . ■

§2.5.4 Countable and Uncountable Sets

Are there more rational or irrational numbers? To answer this question, we need to know in what sense.

Definition 2.5.9 (Cardinality)

A function $f : A \rightarrow B$ is a *bijection* if it is both injective (one-to-one) and surjective (onto). We say that A and B have the same *cardinality*, written

$$|A| = |B|,$$

if there exists a bijection $f : A \rightarrow B$. Thus, cardinality records the size of a set up to a bijection.

Definition 2.5.10

A set S is *finite* if $S = \emptyset$ or if there is a bijection

$$\{1, 2, \dots, n\} \rightarrow S$$

for some $n \in \mathbb{N}$. A set S is *countably infinite* if there is a bijection $f : \mathbb{N} \rightarrow S$.

Example 2.5.11

- (a) The function $f : \mathbb{N} \rightarrow \{2, 4, 6, \dots\}$ defined by $f(n) = 2n$ is a bijection. Thus the set of even natural numbers has the same cardinality as $\mathbb{N}$.
- (b) The function $g : \mathbb{N} \rightarrow \mathbb{Z}$ defined by

$$g(1) = 0, \quad g(2k) = k, \quad g(2k+1) = -k \quad (k \in \mathbb{N})$$

is a bijection. Thus $\mathbb{Z}$ has the same cardinality as $\mathbb{N}$.

Definition 2.5.12

A set S is *countable* if it is finite or countably infinite. A set is *uncountable* if it is not countable.

§3 Sequences and Limits

Concept map. The ϵ -definition of a limit (Theorem 3.1.1) is the hub of the chapter. From it: a limit is unique (Theorem 3.1.5, via Theorem 3.1.4); a convergent sequence is bounded (Theorem 3.1.7); and arithmetic passes through limits (Theorem 3.1.13). Order interacts with limits in Theorems 3.2.1 and 3.2.2, which feed the squeeze theorem (Theorem 3.2.3). The chapter's completeness payoff is the monotone convergence theorem (Theorem 3.2.6): it turns a bounded monotone sequence into a convergent one using the supremum and its ϵ -characterization (Theorem 2.4.4) from §2 — the direct bridge back to the real-number axioms. Its first applications are a recursively defined sequence (Theorem 3.2.7) and Euler's number e (Theorem 3.3.2). Finally, every subsequence of a convergent sequence has the same limit (Theorem 3.3.7), which also gives a test for divergence.

§3.1 Convergent Sequences

The limit definition makes precise what it means for the terms of a sequence to settle near a real number. We use it to distinguish convergence from two ways a sequence can diverge.

§3.1.1 The Limit of a Sequence

A sequence of real numbers is a function $X : \mathbb{N} \rightarrow \mathbb{R}$. We write $x_n = X(n)$ for its n th term and denote the sequence by $\{x_n\}_{n=1}^{\infty}$.

Definition 3.1.1

A sequence $\{x_n\}$ in $\mathbb{R}$ converges to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there is an index $n_\epsilon \in \mathbb{N}$ such that

$$n \geq n_\epsilon \implies |x_n - a| < \epsilon.$$

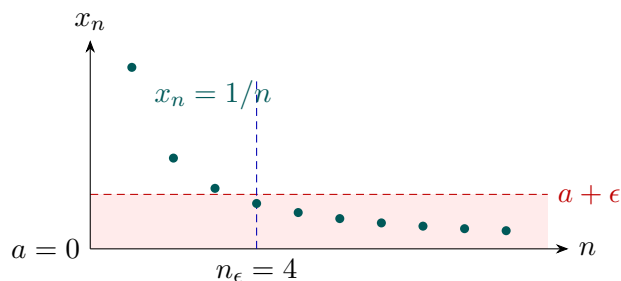
We then write $\lim_{n \rightarrow \infty} x_n = a$.

In symbols,

$$x_n \rightarrow a \iff (\forall \epsilon > 0)(\exists n_\epsilon \in \mathbb{N})(\forall n \geq n_\epsilon) [|x_n - a| < \epsilon].$$

In words: for every tolerance $\epsilon > 0$, all sufficiently late terms lie in $(a - \epsilon, a + \epsilon)$. The index n_ϵ may depend on ϵ , and once $n \geq n_\epsilon$ the inequality must hold for *every* later term, even if earlier terms lie outside the interval.

The points below show $x_n = 1/n$. For the illustrated tolerance $\epsilon = 0.3$, every term from $n_\epsilon = 4$ onward lies in the shaded band around $a = 0$.



For $a \in \mathbb{R}$, the ϵ -neighbourhood of a is

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = (a - \epsilon, a + \epsilon).$$

Thus $x_n \rightarrow a$ precisely when, for every $\epsilon > 0$, there is $n_\epsilon \in \mathbb{N}$ such that $x_n \in V_\epsilon(a)$ for all $n \geq n_\epsilon$.

Example 3.1.2

Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Proof. Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Hence $1/n \rightarrow 0$. ■

Example 3.1.3

Prove that $\lim_{n \rightarrow \infty} \frac{4n+3}{n+1} = 4$.

Scratch work

Simplify the distance first, then see how large n must be:

$$\left| \frac{4n+3}{n+1} - 4 \right| = \left| \frac{-1}{n+1} \right| = \frac{1}{n+1} < \frac{1}{n}.$$

So it is enough that $\frac{1}{n} < \epsilon$. By Archimedes pick n_ϵ with $\frac{1}{n_\epsilon} < \epsilon$; every $n \geq n_\epsilon$ then works.

Proof. Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Therefore the sequence converges to 4. ■

§3.1.2 Uniqueness of the Limit**Proposition 3.1.4**

If $x, y \in \mathbb{R}$ and $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

In symbols

$$(\forall x, y \in \mathbb{R}) \left[(\forall \epsilon > 0) [|x - y| < \epsilon] \Rightarrow x = y \right]$$

Proof. Suppose $x \neq y$. Then $\epsilon = |x - y|/2 > 0$, but $|x - y| > \epsilon$, contradicting the hypothesis. ■

Theorem 3.1.5

A sequence in $\mathbb{R}$ can have at most one limit.

In symbols

$$(\forall a, b \in \mathbb{R}) \left[(x_n \rightarrow a \wedge x_n \rightarrow b) \Rightarrow a = b \right]$$

Scratch work

By Theorem 3.1.4, it suffices to show $|a - b| < \epsilon$ for every $\epsilon > 0$. The only link between a

and b is the sequence, so pass through a term x_n :

$$|a - b| \leq |a - x_n| + |x_n - b|.$$

To make the sum $< \epsilon$, make each piece $< \epsilon/2$: that needs $n \geq n_1$ for the first and $n \geq n_2$ for the second, so take $n \geq \max\{n_1, n_2\}$.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Given $\epsilon > 0$, choose $n_1, n_2 \in \mathbb{N}$ so that $|x_n - a| < \epsilon/2$ whenever $n \geq n_1$ and $|x_n - b| < \epsilon/2$ whenever $n \geq n_2$. For $n \geq \max\{n_1, n_2\}$, the triangle inequality gives

$$|a - b| \leq |a - x_n| + |x_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This holds for every $\epsilon > 0$, so Theorem 3.1.4 gives $a = b$. ■

§3.1.3 Divergence and Boundedness

A sequence diverges if it does not converge to any real number. For example, consider $\{x_n\} = \{0, 1, 0, 1, \dots\}$, with $x_n = 0$ for odd n and $x_n = 1$ for even n .

Negating the limit definition, $\{x_n\}$ diverges precisely when

$$(\forall a \in \mathbb{R})(\exists \epsilon > 0)(\forall N \in \mathbb{N})(\exists n \geq N) [|x_n - a| \geq \epsilon].$$

Both 0 and 1 keep appearing arbitrarily late, so the terms cannot all remain close to one proposed limit.

Suppose $x_n \rightarrow a$ and take $\epsilon = 1/2$. There is an n_ϵ such that $|x_n - a| < 1/2$ for every $n \geq n_\epsilon$. Choose an odd and an even index beyond n_ϵ . The odd term gives $|a| < 1/2$, hence $-1/2 < a < 1/2$. The even term gives $|1 - a| < 1/2$, hence $1/2 < a < 3/2$. These intervals do not overlap, a contradiction. Thus this sequence diverges.

Definition 3.1.6

A sequence $\{x_n\}$ is bounded if there is an $M \in \mathbb{R}$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 3.1.7

Every convergent sequence of real numbers is bounded.

In symbols

$$(\exists a \in \mathbb{R}) [x_n \rightarrow a] \Rightarrow (\exists M \in \mathbb{R})(\forall n \in \mathbb{N}) [|x_n| \leq M]$$

Idea

The tail stays within 1 of the limit; only finitely many terms come before it, and a finite set has a maximum.

Proof. Suppose $x_n \rightarrow a$. Taking $\epsilon = 1$, choose $n_1 \in \mathbb{N}$ so that $|x_n - a| < 1$ for all $n \geq n_1$. For these terms, the triangle inequality gives

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

The remaining terms form a finite set. Thus

$$M = \max\{|x_1|, \dots, |x_{n_1-1}|, 1 + |a|\}$$

bounds every term of the sequence (when $n_1 = 1$, take $M = 1 + |a|$). ■

The converse fails: $\{0, 1, 0, 1, \dots\}$ is bounded but, as shown above, diverges.

Example 3.1.8

The sequences $a_n = (n^2 + 1)/n$ and $b_n = (-1)^n n$ are unbounded.

For $a_n = n + 1/n$, given $M > 0$ choose $n > M$ by the Archimedean Property. Then $|a_n| = n + 1/n > M$. For b_n , the same choice gives $|b_n| = n > M$. In each case no single bound works for every term. Notice that a_n eventually stays positive, while b_n alternates signs; their unboundedness alone does not describe how they diverge.

§3.1.4 Divergence to $\pm\infty$

Definition 3.1.9

A sequence $\{x_n\}$ diverges to $+\infty$, written $\lim_{n \rightarrow \infty} x_n = +\infty$, if for every $M > 0$ there is $n_M \in \mathbb{N}$ such that $x_n > M$ for all $n \geq n_M$.

In symbols, $(\forall M > 0)(\exists n_M \in \mathbb{N})(\forall n \geq n_M)[x_n > M]$.

Definition 3.1.10

A sequence $\{x_n\}$ diverges to $-\infty$, written $\lim_{n \rightarrow \infty} x_n = -\infty$, if for every $M > 0$ there is $n_M \in \mathbb{N}$ such that $x_n < -M$ for all $n \geq n_M$.

In symbols, $(\forall M > 0)(\exists n_M \in \mathbb{N})(\forall n \geq n_M)[x_n < -M]$. We write $\lim_{n \rightarrow \infty} x_n = \pm\infty$, but such a sequence is still divergent: it does not converge to any real number.

Example 3.1.11

Prove that $\lim_{n \rightarrow \infty} \frac{n^2}{1-n} = -\infty$.

Scratch work

For $n > 1$, write $\frac{n^2}{1-n} = -\frac{n^2}{n-1}$. Since $\frac{n^2}{n-1} > n$, it is enough to choose an index after which $n > M$ and $n > 1$.

Proof. Let $M > 0$. By the Archimedean Property choose $n_M \in \mathbb{N}$ with $n_M > \max\{M, 1\}$. For every $n \geq n_M$, we have $n > 1$ and

$$\frac{n^2}{1-n} = -\frac{n^2}{n-1} < -n < -M,$$

because $n^2 > n(n-1)$. Hence the sequence diverges to $-\infty$. ■

Example 3.1.12

Prove that $\lim_{n \rightarrow \infty} (\sqrt{n} + 1) = +\infty$.

Scratch work

To obtain $\sqrt{n} + 1 > M$, it is enough to make $\sqrt{n} > M$, or $n > M^2$. Archimedes supplies such an index.

Proof. Let $M > 0$. Choose $n_M \in \mathbb{N}$ with $n_M > M^2$ by the Archimedean Property. If $n \geq n_M$, then $\sqrt{n} \geq \sqrt{n_M} > M$, so $\sqrt{n} + 1 > M$. Thus the sequence diverges to $+\infty$. ■

§3.1.5 Elementary Limit Laws

The limit definition lets us pass to limits through arithmetic. For a quotient, the denominator's limit must be nonzero.

Theorem 3.1.13

If $\lim_{n \rightarrow \infty} x_n = a$ and $\lim_{n \rightarrow \infty} y_n = b$ in $\mathbb{R}$, then

- (i) $\lim_{n \rightarrow \infty} c = c$ for every constant $c \in \mathbb{R}$;
- (ii) $\lim_{n \rightarrow \infty} (cx_n) = ca$ for every $c \in \mathbb{R}$;
- (iii) $\lim_{n \rightarrow \infty} (x_n + y_n) = a + b$;
- (iv) $\lim_{n \rightarrow \infty} (x_n - y_n) = a - b$;
- (v) $\lim_{n \rightarrow \infty} (x_n y_n) = ab$;
- (vi) if $b \neq 0$ and $y_n \neq 0$ for every n , then $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{a}{b}$;
- (vii) $\lim_{n \rightarrow \infty} |x_n| = |a|$;
- (viii) $\lim_{n \rightarrow \infty} x_n^k = a^k$ for every positive integer k .

In symbols

- (i) $(\forall c \in \mathbb{R}) \left[\lim_{n \rightarrow \infty} c = c \right]$,
- (ii) $(\forall c \in \mathbb{R}) \left[\lim_{n \rightarrow \infty} (cx_n) = ca \right]$,
- (iii) $\lim_{n \rightarrow \infty} (x_n + y_n) = a + b$,
- (iv) $\lim_{n \rightarrow \infty} (x_n - y_n) = a - b$,
- (v) $\lim_{n \rightarrow \infty} (x_n y_n) = ab$,
- (vi) $b \neq 0 \wedge (\forall n \in \mathbb{N}) [y_n \neq 0] \Rightarrow \lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{a}{b}$,
- (vii) $\lim_{n \rightarrow \infty} |x_n| = |a|$,
- (viii) $(\forall k \in \mathbb{N}) \left[\lim_{n \rightarrow \infty} x_n^k = a^k \right]$.

Proof. For (i), let $\epsilon > 0$. Every term of the constant sequence satisfies $|c - c| = 0 < \epsilon$, so any $n_\epsilon \in \mathbb{N}$ works.

For (ii), fix $c \in \mathbb{R}$. If $c = 0$, then $cx_n = ca = 0$ for every n . If $c \neq 0$, given $\epsilon > 0$ use $x_n \rightarrow a$ with tolerance $\epsilon/|c|$ to choose n_ϵ such that $|x_n - a| < \epsilon/|c|$ for all $n \geq n_\epsilon$. Hence

$$|cx_n - ca| = |c| |x_n - a| < \epsilon$$

for all such n , as required.

For (iii), given $\epsilon > 0$, choose n late enough that both $|x_n - a| < \epsilon/2$ and $|y_n - b| < \epsilon/2$. Then

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b| < \epsilon.$$

For (iv), apply (ii) with $c = -1$ and then (iii).

For (v), convergent sequences are bounded, so choose $M > 0$ with $|x_n| \leq M$ for every n . Given $\epsilon > 0$, take n late enough that $|y_n - b| < \epsilon/(2M)$ and $|x_n - a| < \epsilon/(2(|b| + 1))$.

Thus

$$|x_n y_n - ab| \leq |x_n| |y_n - b| + |b| |x_n - a| < \epsilon.$$

For (vi), we first prove $1/y_n \rightarrow 1/b$. Since $b \neq 0$, eventually $|y_n - b| < |b|/2$, which gives $|y_n| > |b|/2$. Given $\epsilon > 0$, take n still later so that $|y_n - b| < \epsilon |b|^2/2$. Then

$$\left| \frac{1}{y_n} - \frac{1}{b} \right| = \frac{|y_n - b|}{|y_n| |b|} < \frac{2 |y_n - b|}{|b|^2} < \epsilon.$$

Apply (v) to x_n and $1/y_n$.

For (vii), the reverse triangle inequality gives $||x_n| - |a|| \leq |x_n - a|$. For (viii), use induction: $x_n^1 \rightarrow a$ by hypothesis, and if $x_n^k \rightarrow a^k$, then (v) gives $x_n^{k+1} = x_n^k x_n \rightarrow a^k a = a^{k+1}$. ■

Theorem 3.1.14

If $x_n \rightarrow +\infty$ and $y_n \rightarrow L > 0$, then $x_n y_n \rightarrow +\infty$.

In symbols

$$[x_n \rightarrow +\infty \wedge y_n \rightarrow L \wedge L > 0] \Rightarrow x_n y_n \rightarrow +\infty$$

Idea

Eventually y_n stays above the positive number $L/2$. Make x_n large enough that $(L/2)x_n$ exceeds the requested bound.

Proof. Let $M > 0$. Since $y_n \rightarrow L$, there is n_1 such that $y_n > L/2$ for every $n \geq n_1$. Since $x_n \rightarrow +\infty$, there is n_2 such that $x_n > 2M/L > 0$ for every $n \geq n_2$. Thus, if $n \geq \max\{n_1, n_2\}$,

$$x_n y_n > \frac{2M}{L} \cdot \frac{L}{2} = M.$$

Hence $x_n y_n \rightarrow +\infty$. ■

Example 3.1.15

Evaluate $\lim_{n \rightarrow \infty} \frac{n^2+2}{n-1}$.

Proof. For $n > 1$,

$$\frac{n^2+2}{n-1} = n \frac{1+2/n^2}{1-1/n}.$$

By the limit laws and $1/n \rightarrow 0$, the second factor tends to 1. Theorem 3.1.14 then shows that the original sequence diverges to $+\infty$. ■

Example 3.1.16

Evaluate $\lim_{n \rightarrow \infty} \frac{n^3+5}{n+1}$.

Proof. For $n \geq 1$,

$$\frac{n^3+5}{n+1} = n^2 \frac{1+5/n^3}{1+1/n}.$$

The limit laws give $\frac{1+5/n^3}{1+1/n} \rightarrow 1$. Also $n^2 \rightarrow +\infty$, since $n^2 \geq n$ and the natural numbers are unbounded above. By Theorem 3.1.14, the quotient diverges to $+\infty$. ■

In practice

- **Prove convergence.** Simplify $|x_n - a| < \epsilon$, then choose an index that makes it hold for every later term.
- **Prove divergence to infinity.** Given $M > 0$, find a lower bound on n that forces $x_n > M$ or $x_n < -M$.
- **Use the limit laws.** Factor a rational expression into a power of n and a factor with a finite nonzero limit. Check whether the power converges or diverges before taking the limit.

§3.2 Limits, Order, and Monotone Sequences

Limits preserve order, and completeness turns a bounded monotone sequence into a convergent one. These facts let us find limits even when the limit laws alone do not apply.

§3.2.1 Limits and Inequalities**Theorem 3.2.1**

If $\lim_{n \rightarrow \infty} x_n = a$ and $x_n \geq 0$ for every $n \in \mathbb{N}$, then $a \geq 0$.

In symbols

$$\left[\lim_{n \rightarrow \infty} x_n = a \wedge (\forall n \in \mathbb{N}) [x_n \geq 0] \right] \Rightarrow a \geq 0$$

Idea

If $a < 0$, convergence would force every sufficiently late term into a small interval around a that lies below zero.

Proof. Suppose instead that $a < 0$. Take $\epsilon = -a/2 > 0$. Since $\lim_{n \rightarrow \infty} x_n = a$, some $n_\epsilon \in \mathbb{N}$ satisfies $|x_n - a| < \epsilon$ whenever $n \geq n_\epsilon$. For such n ,

$$x_n < a + \epsilon = \frac{a}{2} < 0,$$

contradicting $x_n \geq 0$. Thus $a \geq 0$. ■

The same conclusion holds if $x_n \geq 0$ only for all sufficiently large n : the proof uses just one term after both thresholds.

Theorem 3.2.2

If $\lim_{n \rightarrow \infty} x_n = a$, $\lim_{n \rightarrow \infty} y_n = b$, and $x_n \leq y_n$ for every $n \in \mathbb{N}$, then $a \leq b$.

In symbols

$$\left[\lim_{n \rightarrow \infty} x_n = a \wedge \lim_{n \rightarrow \infty} y_n = b \wedge (\forall n \in \mathbb{N}) [x_n \leq y_n] \right] \Rightarrow a \leq b$$

Proof. Set $z_n = y_n - x_n \geq 0$. The difference law (Theorem 3.1.13, part (iv)) gives $\lim_{n \rightarrow \infty} z_n = b - a$. By Theorem 3.2.1, $b - a \geq 0$, so $a \leq b$. ■

Strict inequalities are not preserved: $\frac{1}{n} > 0$ for every $n \in \mathbb{N}$, yet $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. From $x_n < y_n$ for all n we may conclude only $a \leq b$.

§3.2.2 The Squeeze Theorem

Theorem 3.2.3 (Squeeze Theorem)

Suppose $x_n \leq y_n \leq z_n$ for every $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} x_n = a$, $\lim_{n \rightarrow \infty} z_n = a$. Then $\lim_{n \rightarrow \infty} y_n = a$.

In symbols

$$[(\forall n \in \mathbb{N}) [x_n \leq y_n \leq z_n] \wedge \lim_{n \rightarrow \infty} x_n = a \wedge \lim_{n \rightarrow \infty} z_n = a] \Rightarrow \lim_{n \rightarrow \infty} y_n = a$$

Proof. Let $\epsilon > 0$. Choose $n_1, n_2 \in \mathbb{N}$ such that $|x_n - a| < \epsilon$ for $n \geq n_1$ and $|z_n - a| < \epsilon$ for $n \geq n_2$. If $n \geq \max\{n_1, n_2\}$, then

$$a - \epsilon < x_n \leq y_n \leq z_n < a + \epsilon.$$

Thus $|y_n - a| < \epsilon$ for every such n , so $\lim_{n \rightarrow \infty} y_n = a$. ■

Example 3.2.4

Prove that $\lim_{n \rightarrow \infty} \frac{\sin n}{\sqrt{n}} = 0$.

Proof. Since $-1 \leq \sin n \leq 1$ and $\sqrt{n} > 0$,

$$-\frac{1}{\sqrt{n}} \leq \frac{\sin n}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

The outer sequences tend to 0: given $\epsilon > 0$, the Archimedean Property (Theorem 2.5.3) supplies $n_\epsilon \in \mathbb{N}$ with $n_\epsilon > 1/\epsilon^2$, and every $n \geq n_\epsilon$ gives $1/\sqrt{n} \leq 1/\sqrt{n_\epsilon} < \epsilon$; the lower bound $-1/\sqrt{n}$ tends to 0 by the same n_ϵ . Theorem 3.2.3 now gives the result. ■

§3.2.3 Monotone Sequences**Definition 3.2.5**

A sequence $\{x_n\}$ is *increasing* if $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$, and *decreasing* if $x_n \geq x_{n+1}$ for every $n \in \mathbb{N}$. It is *monotone* if it is increasing or decreasing.

Theorem 3.2.6 (Monotone Convergence Theorem)

A monotone sequence of real numbers converges if and only if it is bounded. More precisely:

(i) If $\{x_n\}$ is bounded and increasing, then

$$\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\}.$$

(ii) If $\{x_n\}$ is bounded and decreasing, then

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\}.$$

In symbols

$$\begin{aligned} & [\{x_n\} \text{ monotone}] \\ & \Rightarrow [(\exists a \in \mathbb{R}) [\lim_{n \rightarrow \infty} x_n = a] \iff \{x_n\} \text{ bounded}], \\ & \text{(i) } [\{x_n\} \text{ bounded and increasing}] \\ & \quad \Rightarrow \lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\}, \\ & \text{(ii) } [\{x_n\} \text{ bounded and decreasing}] \\ & \quad \Rightarrow \lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\}. \end{aligned}$$

Idea

Completeness gives a supremum or infimum. A term lies within ϵ of that bound, and monotonicity keeps every later term there.

Proof. Every convergent sequence is bounded (Theorem 3.1.7), proving the forward implication. For the converse, (i) and (ii) cover the two kinds of monotone sequence.

For (i), suppose $\{x_n\}$ is bounded and increasing. The set $S = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above, so completeness gives $s = \sup S$. Let $\epsilon > 0$. The ϵ -characterization of the supremum (Theorem 2.4.4) gives an index n_ϵ with $s - \epsilon < x_{n_\epsilon}$. For every $n \geq n_\epsilon$, monotonicity and the upper-bound property give

$$s - \epsilon < x_{n_\epsilon} \leq x_n \leq s < s + \epsilon.$$

Hence $\lim_{n \rightarrow \infty} x_n = s$.

For (ii), suppose $\{x_n\}$ is bounded and decreasing. The set S is nonempty and bounded below, so $t = \inf S$ exists (Theorem 2.5.1). Let $\epsilon > 0$. Since $t + \epsilon > t$ and t is the *greatest* lower bound, $t + \epsilon$ is not a lower bound of S ; so there is an index n_ϵ with $x_{n_\epsilon} < t + \epsilon$. For $n \geq n_\epsilon$, monotonicity and the lower-bound property give

$$t - \epsilon < t \leq x_n \leq x_{n_\epsilon} < t + \epsilon.$$

Thus $\lim_{n \rightarrow \infty} x_n = t$. ■

§3.2.4 A Recursively Defined Sequence

The Monotone Convergence Theorem is most useful for sequences defined by recursion, where no formula for x_n is available. It shows that a limit exists; the recursion then determines its value.

Example 3.2.7

Let $x_1 = 1$ and $x_{n+1} = \frac{2x_n+3}{4}$ for all $n \in \mathbb{N}$. Show that $\{x_n\}$ converges and find its limit.

The first terms are

$$x_1 = 1, \quad x_2 = \frac{2 \cdot 1 + 3}{4} = \frac{5}{4} = 1.25, \quad x_3 = \frac{2 \cdot \frac{5}{4} + 3}{4} = \frac{11}{8} = 1.375,$$

which suggests that the sequence increases and stays below 2.

Idea

Prove by induction that the sequence is bounded above and increasing; the Monotone Convergence Theorem then gives a limit a , and letting $n \rightarrow \infty$ in the recursion gives an equation for a .

Proof. Step 1: $x_n < 2$ for all $n \in \mathbb{N}$. We use induction on n . For $n = 1$, $x_1 = 1 < 2$. Let $k \in \mathbb{N}$ and assume $x_k < 2$. Then

$$2x_k < 4 \implies 2x_k + 3 < 7 \implies x_{k+1} = \frac{2x_k + 3}{4} < \frac{7}{4} < 2.$$

By induction, $x_n < 2$ for every $n \in \mathbb{N}$.

Step 2: $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$. Again by induction. For $n = 1$, $x_1 = 1 < \frac{5}{4} = x_2$. Let $k \in \mathbb{N}$ and assume $x_k \leq x_{k+1}$. Then

$$2x_k + 3 \leq 2x_{k+1} + 3 \implies x_{k+1} = \frac{2x_k + 3}{4} \leq \frac{2x_{k+1} + 3}{4} = x_{k+2}.$$

By induction, $\{x_n\}$ is increasing.

Step 3: the limit. By Step 2, $x_n \geq x_1 = 1$ for every n , and by Step 1, $x_n < 2$; so $\{x_n\}$ is bounded and increasing. By Theorem 3.2.6 there is $a \in \mathbb{R}$ with $\lim_{n \rightarrow \infty} x_n = a$.

The shifted sequence also satisfies $\lim_{n \rightarrow \infty} x_{n+1} = a$: if $|x_n - a| < \epsilon$ for all $n \geq n_\epsilon$, then also $|x_{n+1} - a| < \epsilon$ for all $n \geq n_\epsilon$, since $n + 1 \geq n_\epsilon$. On the other hand, $x_{n+1} = \frac{1}{2}x_n + \frac{3}{4}$, so the limit laws (Theorem 3.1.13, parts (i)–(iii)) give

$$\lim_{n \rightarrow \infty} x_{n+1} = \frac{1}{2}a + \frac{3}{4} = \frac{2a + 3}{4}.$$

A limit is unique (Theorem 3.1.5), so

$$a = \frac{2a + 3}{4} \implies 4a = 2a + 3 \implies a = \frac{3}{2}.$$

Hence $\lim_{n \rightarrow \infty} x_n = \frac{3}{2}$. ■

Existence must come first. The equation $a = \frac{2a+3}{4}$ only says what the limit would be if there is one: for $x_1 = 1$ and $x_{n+1} = 2x_n$, the same manipulation gives $a = 2a$, so $a = 0$, yet $x_n = 2^{n-1}$ diverges to $+\infty$. The bound 2 in Step 1 is not special; any upper bound at least $\frac{3}{2}$ (the solution of the limit equation) would work.

In practice

- **Use the squeeze theorem.** Bound a difficult sequence above and below by sequences with the same known limit; verify that both bounds hold for all sufficiently large indices.
- **Use monotonicity.** Prove the direction of monotonicity and find a bound. Completeness then gives the limit as the supremum or infimum of the terms.

- **Recursive sequences.** Compute a few terms to guess the direction and a bound; prove both by induction; apply the Monotone Convergence Theorem; then pass to the limit on both sides of the recursion and solve for the limit.

§3.3 Euler's Number and Subsequences

The Monotone Convergence Theorem proves that $(1 + \frac{1}{n})^n$ converges without knowing its limit in advance; that limit defines e . We then pass from a sequence to its subsequences, which inherit its limit.

§3.3.1 Euler's Number

Consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n \quad (n \in \mathbb{N}).$$

Neither the base nor the exponent is fixed, so the limit laws do not apply: the base tends to 1 while the exponent grows. We expand the power instead.

Theorem 3.3.1 (Binomial Theorem)

For every $x \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!} x^2 + \cdots + x^n,$$

where $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!}$.

In symbols

$$(\forall x \in \mathbb{R})(\forall n \in \mathbb{N}) \left[(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k \right]$$

The lecture used this without proof.

With $x = \frac{1}{n}$, the term of index k is

$$\binom{n}{k} \frac{1}{n^k} = \frac{1}{k!} \cdot \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-k+1}{n} = \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Hence

$$\begin{aligned} x_n &= 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{2!} \cdot \frac{1}{n^2} + \cdots + \frac{n(n-1)\cdots 1}{n!} \cdot \frac{1}{n^n} \\ &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \cdots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right). \end{aligned}$$

Every bracket lies in $(0, 1]$, which gives two comparisons.

Against a simpler sum. Dropping the brackets can only increase each term, so

$$x_n \leq y_n, \quad \text{where } y_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

Against the next term of the sequence. Replacing n by $n + 1$,

$$\begin{aligned} x_{n+1} &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n+1}\right) + \frac{1}{3!} \left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) + \cdots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n+1}\right) \cdots \left(1 - \frac{n-1}{n+1}\right) \\ &\quad + \frac{1}{(n+1)!} \left(1 - \frac{1}{n+1}\right) \cdots \left(1 - \frac{n}{n+1}\right). \end{aligned}$$

Since $1 - \frac{j}{n} < 1 - \frac{j}{n+1}$ for $1 \leq j \leq n-1$, each of the first $n+1$ terms of x_{n+1} is at least the corresponding term of x_n , and x_{n+1} has one more positive term. So $x_n < x_{n+1}$.

Proposition 3.3.2

The sequence $x_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and satisfies $2 \leq x_n < 3$ for every $n \in \mathbb{N}$. Hence it converges.

In symbols

$$(\forall n \in \mathbb{N}) [x_n < x_{n+1} \wedge 2 \leq x_n < 3] \wedge (\exists e \in \mathbb{R}) \left[\lim_{n \rightarrow \infty} x_n = e \right]$$

Definition 3.3.3 (Euler's Number)

The limit $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ is called *Euler's number*.

Idea

The comparisons above give monotonicity and $x_n \leq y_n$. Bounding $\frac{1}{k!}$ by a power of $\frac{1}{2}$ turns y_n into a geometric sum below 3; the Monotone Convergence Theorem finishes.

Proof. Step 1: monotone. The term-by-term comparison above shows $x_n < x_{n+1}$ for every n .

Step 2: $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$ for $k \in \mathbb{N}$. Each of the $k-1$ factors $2, 3, \dots, k$ of $k!$ is at least 2, so

$$\frac{1}{k!} = \frac{1}{1 \cdot 2 \cdots k} \leq \frac{1}{1} \cdot \frac{1}{2} \cdots \frac{1}{2} = \frac{1}{2^{k-1}}.$$

Step 3: bounds. By Step 2 and the finite geometric sum,

$$y_n = 1 + \sum_{k=1}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \left(\frac{1}{2}\right)^{k-1} = 1 + \frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} = 3 - \frac{1}{2^{n-1}} < 3.$$

Since $x_1 = 2$ and the sequence increases, $2 \leq x_n \leq y_n < 3$ for every n .

Step 4: convergence. $\{x_n\}$ is bounded and increasing, so by Theorem 3.2.6 it converges. ■

The lecture bounded y_n by the full geometric series $1 + \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{k-1} = 3$; the finite sum above gives the same bound without infinite series, which the course has not defined yet. Theorem 3.3.2 tells us only that e exists and, by Theorem 3.2.2, that $2 \leq e \leq 3$; its value 2.718... needs more work.

§3.3.2 Subsequences

Definition 3.3.4 (Subsequence)

Let $\{x_n\}$ be a sequence of real numbers and let $n_1 < n_2 < n_3 < \dots$ be a strictly increasing sequence of natural numbers. Then

$$\{x_{n_k}\} = \{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$$

is called a *subsequence* of $\{x_n\}$.

In words: keep infinitely many terms, in their original order, and discard the rest. The index of the subsequence is k ; n_k records which term of the original sequence is kept.

Example 3.3.5

Let $\{x_n\} = \{1, -1, 1, -1, \dots\}$, that is, $x_n = (-1)^{n+1}$.

The sequence itself has no limit, but choosing indices well produces convergent subsequences. The odd indices $n_k = 2k - 1$ give

$$\{x_{2k-1}\} = \{1, 1, 1, \dots\},$$

which converges to 1, and the even indices $n_k = 2k$ give

$$\{x_{2k}\} = \{-1, -1, -1, \dots\},$$

which converges to -1 .

Lemma 3.3.6

If $n_1 < n_2 < n_3 < \dots$ are natural numbers, then $k \leq n_k$ for every $k \in \mathbb{N}$.

In symbols

$$(\forall k \in \mathbb{N}) [n_k < n_{k+1}] \Rightarrow (\forall k \in \mathbb{N}) [k \leq n_k]$$

Proof. By induction on k . For $k = 1$, $n_1 \in \mathbb{N}$ gives $1 \leq n_1$. If $k \leq n_k$, then, since $n_k < n_{k+1}$ are integers,

$$k + 1 \leq n_k + 1 \leq n_{k+1}.$$

■

Theorem 3.3.7

If a sequence $\{x_n\}$ of real numbers converges to a real number x , then every subsequence of $\{x_n\}$ converges to x .

In symbols

$$\left[\lim_{n \rightarrow \infty} x_n = x \wedge (\forall k \in \mathbb{N}) [n_k < n_{k+1}] \right] \Rightarrow \lim_{k \rightarrow \infty} x_{n_k} = x$$

Idea

The k -th term of the subsequence sits at position $n_k \geq k$ of the original sequence, so it is at least as late; the same threshold n_ϵ works.

Proof. Let $\{x_{n_k}\}$ be a subsequence and let $\epsilon > 0$. Since $\lim_{n \rightarrow \infty} x_n = x$, there is $n_\epsilon \in \mathbb{N}$ such that

$$n \geq n_\epsilon \implies |x_n - x| < \epsilon.$$

By Theorem 3.3.6, $n_k \geq k$ for every k . Hence

$$k \geq n_\epsilon \implies n_k \geq k \geq n_\epsilon \implies |x_{n_k} - x| < \epsilon.$$

Thus $\lim_{k \rightarrow \infty} x_{n_k} = x$. ■

Read contrapositively, Theorem 3.3.7 is a divergence test: if two subsequences converge to different limits, the sequence diverges. In Theorem 3.3.5, the subsequences tend to 1 and -1 , so $\{(-1)^{n+1}\}$ diverges.

In practice

- **Expand, then compare.** When the limit laws do not apply, expand the term (here with the binomial theorem) and compare the expansion term by term: with a simpler sum for a bound, and with the next term of the sequence for monotonicity.
- **Bound $\frac{1}{k!}$ by $\frac{1}{2^{k-1}}$.** This turns a factorial sum into a geometric sum with an explicit bound.
- **Divergence by subsequences.** To show a sequence diverges, find two subsequences with different limits.